序列期末報告

第四組 黃琳鈞、胡郡珍、李姿穎

大綱

- PACP (L = odd)
  - g_0 、 g_1 的限制
  - Legendre sequence
- Hamming Distance
- PQCP
  - Calculate w_a, w_b
- Flow Chart
- Result

PACP (L = odd)的g_0、 g_1的限制(1/3)

$$A(x)A(x^{-1}) + B(x)B(x^{-1}) = \sum_{u=0}^{L-1}[\rho(a;u) + \rho(b;u)] \cdot x^u \quad , A(x) = \sum_{i=0}^{L-1} a_i x^i$$

- 左式=> x=1代入: $|A(1)|^2 + |B(1)|^2 = [g_0 \cdot 1 + (L - g_0) \cdot (-1)] + [g_1 \cdot 1 + (L - g_1) \cdot (-1)]$

$$= [(2g_0 - L)^2 + (2g_1 - L)^2]$$

$$= 2\,[(g_0 - g_1)^2 + (L - g_0 - g_1)^2]$$

- 右式=> x=1代入: $\sum_{u=0}^{L-1}[\rho(a;u) + \rho(b;u)] \cdot 1^u = \sum_{u=0}^{L-1} c(u) \cdot 1^u$

$$= 2L + \sum_{u=1}^{\frac{L-1}{2}}[c(u) + c(L-u)]$$

$$= 2L + \sum_{u=1}^{\frac{L-1}{2}} 2\,c(u)$$

$$= 2L + \sum_{u=1}^{\frac{L-1}{2}} 2 \cdot \{\pm 2\}$$

$$= 2L + \sum_{u=1}^{\frac{L-1}{2}} 4 \cdot \{\pm 1\}$$

PACP (L = odd)的g_0、 g_1的限制(2/3)

- 左式=右式:

$$2\,[(g_0-g_1)^2+(L-g_0-g_1)^2] \;=2L+\sum_{u=1}^{\frac{L-1}{2}} 4\cdot\{\pm 1\}$$

$$=>\;[(g_0-g_1)^2+(L-g_0-g_1)^2]=L+\underline{\sum_{u=1}^{\frac{L-1}{2}} 2\cdot\{\pm 1\}}$$

$$-\frac{L-1}{2}\le\sum_{u=1}^{\frac{L-1}{2}}\{\pm 1\}\le\frac{L-1}{2}$$

$$\boxed{1\le(g_0-g_1)^2+(L-g_0-g_1)^2\le 2L-1}$$

PACP (L = odd)的g_0、 g_1的限制(3/3)

$$(g_0 - g_1)^2 + (L - g_0 - g_1)^2 = L + 2\boxed{\sum_{u=1}^{\frac{L-1}{2}}\{\pm 1\}}$$

總共有$\frac{L-1}{2}$項，每一樣元素$\in \{\pm 1\}$
元素若從$+1 \rightarrow -1$，差值為2

$$\sum_{u=1}^{\frac{L-1}{2}}\{\pm 1\} = \frac{L-1}{2}, \frac{L-1}{2} - 2, \frac{L-1}{2} - 4, \ldots$$

$$(g_0 - g_1)^2 + (L - g_0 - g_1)^2 = \{\, 2L - 1\,, 2L - 5\,, \ldots, 1\,\}$$

$$\boxed{2\mathrm{L} - 1 - [(g_0 - g_1)^2 + (L - g_0 - g_1)^2] \equiv 0 \pmod 4}$$

Legendre sequence

- Length of Legendre sequence must satisfy : $1. p \epsilon prime \quad 2. p(mod4) \equiv 3$
- Definition of Legendre symbol

$$\left(\frac{n}{P}\right) = \begin{cases} 1, if\ n\ is\ quadratic\ residue \\ 0, if\ n \equiv 0 \\ -1, otherwise \end{cases}$$

- Euler's criterion : for p is an odd prime,

  if $\gcd(n,p) = 1$, then $\left(\frac{n}{P}\right) \equiv n^{\frac{p-1}{2}} (\bmod p) = \{ {}^{1\ , residue}_{-1, non-residue}$

- Known property of Legendre sequences :

$$\sum_{u=0}^{L-1} X(n)\ X(n+u) = -1, \quad u \neq 0$$

Assume L = 7

$1^2 \equiv 1$, $2^2 \equiv 4$, $3^2 \equiv 2$, $4^2 \equiv 2$, $5^2 \equiv 4$, $6^2 \equiv 1$

{1,2,4}→這些位置填1

{3,5,6}→這些位置填-1

{0} → assume 1

生成的序列 A = {1,1,1,-1,1,-1,-1}

Hamming Distance (L is odd)

- 比較兩個序列有多少個位置不同
- 用來衡量序列與其位移後的差異，進而計算autocorrelation
- $\rho(a;u) + \rho(b;u) <=> d_A(u) + d_B(u)$

$d_A(u) \Rightarrow$ a和a循環位移u格後的Hamming distance

$$\Rightarrow d_A(u) = \text{popcount}\left(\text{a} \oplus \text{rot}^u(\text{a})\right)$$

- $\rho(a;u) = \sum_{i=0}^{L-1} a_i a_{(i+u)\text{modL}}$

$$= [L - d_A(u)] \cdot 1 \ + \ d_A(u) \cdot (-1)$$

$$= L - 2d_A(u)$$

- $\rho(b;u) = L - 2d_B(u)$

- $|\rho(a;u) + \rho(b;u)| = 2, \qquad u \neq 0$

$$=> |\ 2L \ - \ 2d_A(u) - \ 2d_\text{B}(u)| \ = \ 2$$

$$=> d_A\ (u) + d_B(u) = \ L \ \pm 1$$

$$\boxed{d_A(u) + d_B(u) = \ L \ \pm 1}$$

Hamming Distance (L is even)

- $|\rho(a;u) + \rho(b;u)| = 4, \qquad u = \frac{L}{2}$

$$=> |\ (L - 2d_A\left(\frac{L}{2}\right)) + (L - 2d_B\left(\frac{L}{2}\right))\ | = 4$$

$$=> |\ 2L - 2d_A\left(\frac{L}{2}\right) - 2d_B\left(\frac{L}{2}\right)\ | = 4$$

$$=> d_A\left(\frac{L}{2}\right) + d_B\left(\frac{L}{2}\right) = L \pm 2$$

$$\boxed{d_A\left(\frac{L}{2}\right) + d_B\left(\frac{L}{2}\right) = L \pm 2}$$

- $\rho(a;u) + \rho(b;u) = 0, \qquad u \neq \frac{L}{2}, 0$

$$=> (L - 2d_A(u)) + (L - 2d_B(u)) = 0$$

$$=> 2L - 2d_A(u) - 2d_B(u) = 0$$

$$=> d_A(u) + d_B(u) = L$$

$$\boxed{d_A(u) + d_B(u) = L}$$

PQCP : calculate w_a, w_b

$$\sum_{u=0}^{L-1}[\rho(a;u)+\rho(b;u)] = \left(\sum_i a_i\right)^2 + \left(\sum_i b_i\right)^2 = P_a^2 + P_b^2$$

Case1. 有兩根+4 , $P_a^2 + P_b^2 = \underbrace{2L}_{u=0} + 8$

Case2. 有兩根 − 4 , $P_a^2 + P_b^2 = 2L - 8$

	44	46	58	68	86	90	94
Case1	X	6^2+8^2	X	0^2+12^2	6^2+12^2	X	0^2+14^2
Case2	4^2+8^2	X	X	8^2+8^2	8^2+10^2	X	6^2+12^2

$P_a = 1\cdot(L-w_a) + (-1)w_a = L - 2w_a$
$P_b = L - 2w_b$

L = 46
46 - 2 w_a = 6 $\rightarrow w_a = 20$
46 - 2 w_a = -6 $\rightarrow w_a = 26$
46 - 2 w_b = 8 $\rightarrow w_b = 19$
46 - 2 w_b = -8 $\rightarrow w_b = 27$
w_a =(20,26) , w_b = (19,27)

Flow Chart (Odd)

$$2L - 1 - [(g_0 - g_1)^2 + (L - g_0 - g_1)^2] \equiv 0 \pmod 4$$

E.g.

L = 5

1	2	3	4	5	6	7	8	9

k=0
C=0000

1	0	1	1	5	1	1	0	1

k=2
C=0101

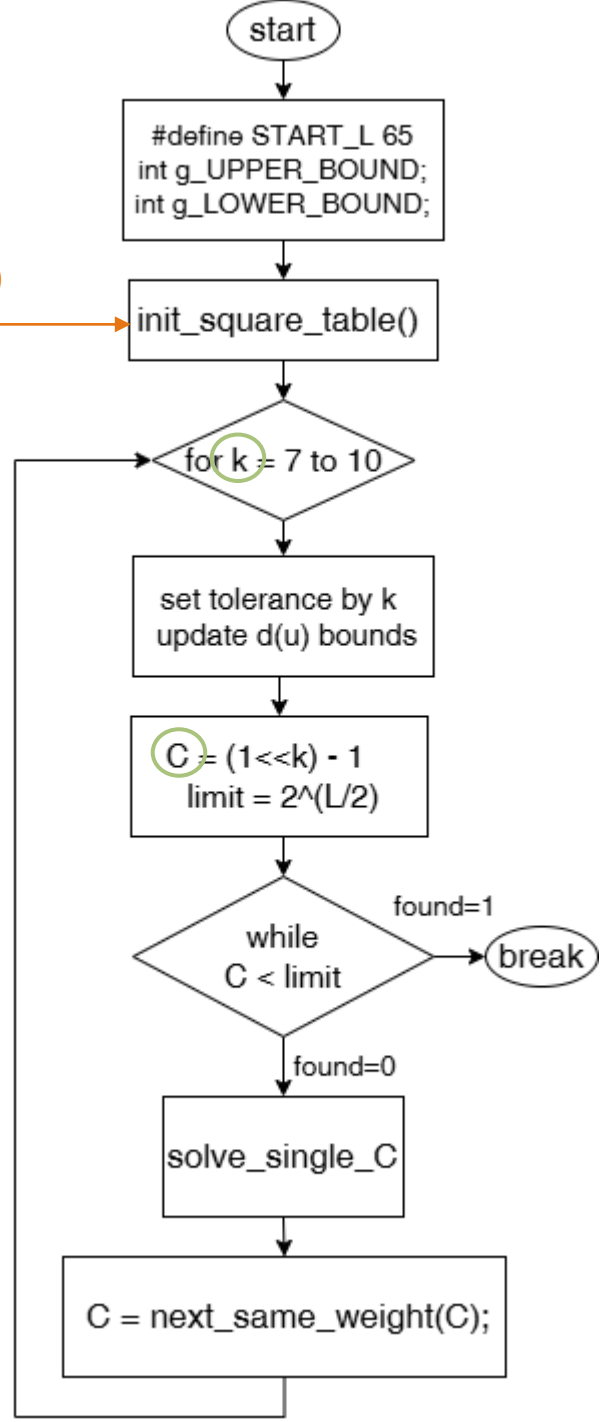
1	0	1	1	5	1	1	1	0

k=4
C=1111

1	0	1	1	5	0	0	1	0

L = 27 k=2, C=2=10

A=101011011110000011110110110

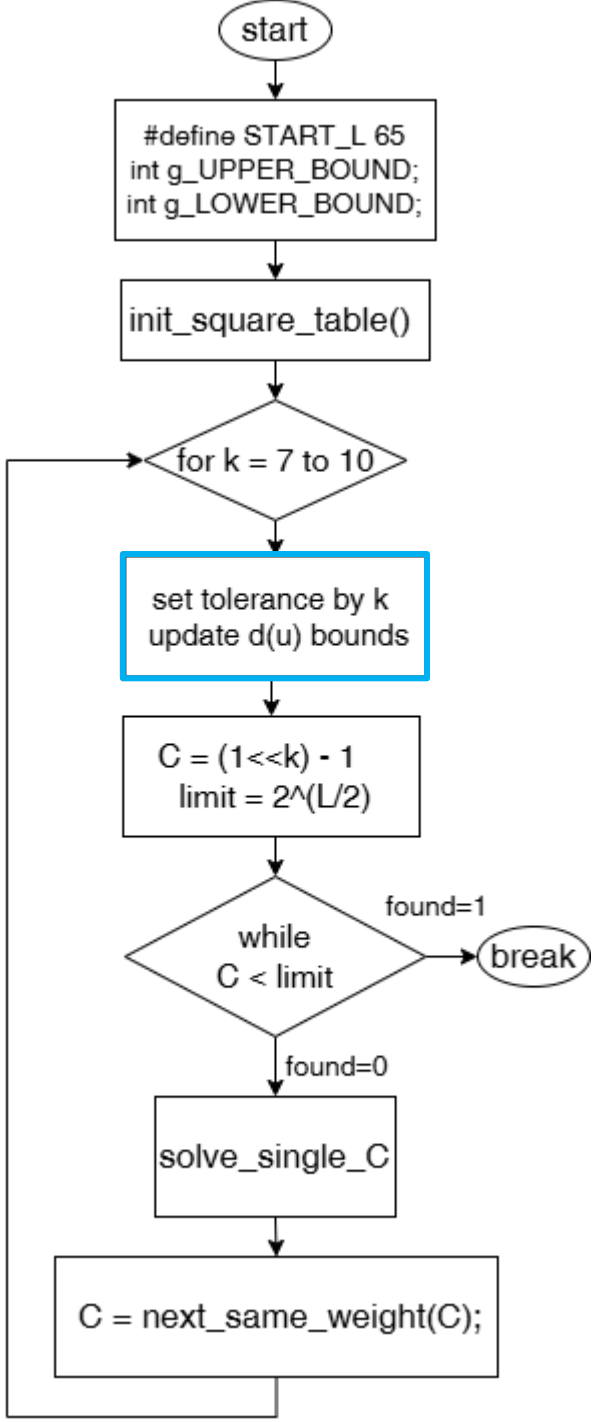


Flow Chart (Odd)

隨著k、L變大，correlation bound調整

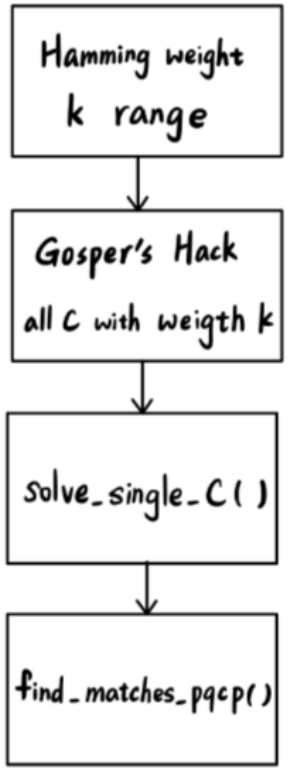
```
for (int k = 0; k <= 10; k++)
{
    if (g_found_any)
        break;

    int tolerance;
    if (k <= 5)
        tolerance = 3;

    else
        tolerance = 2;

    g_UPPER_BOUND = g_L / 2 + tolerance;
    g_LOWER_BOUND = (g_L / 2 - tolerance < 0)? 0 : g_L / 2 - tolerance;
```

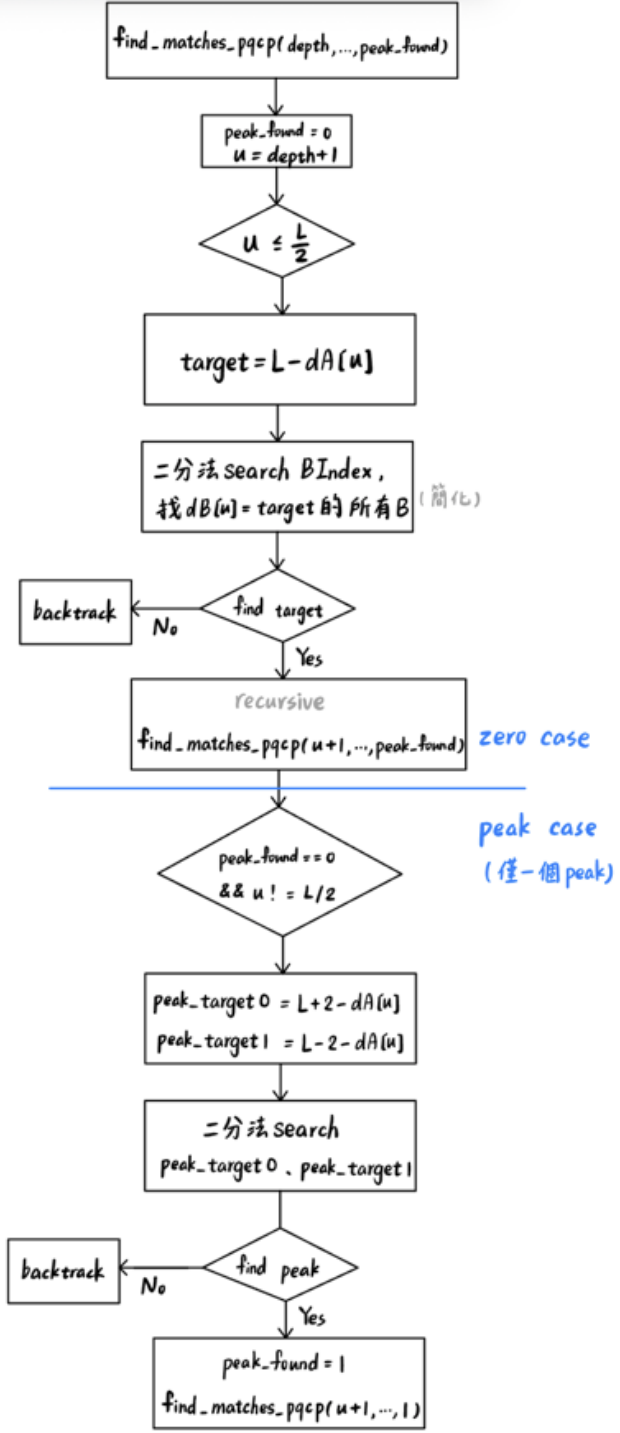
L=31 bound=±2 k=0~20 都無解
L=31 bound=±3 k=2 即有解



Flow Chart (PQCP)

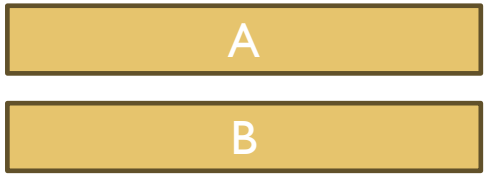
Main



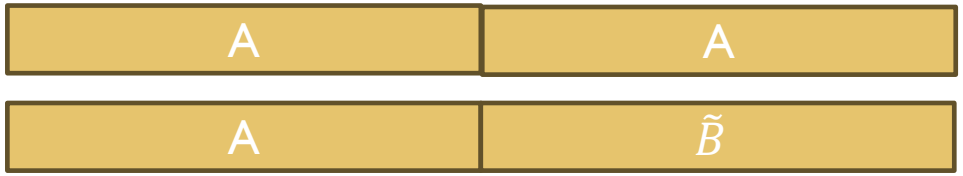


Search Odd Length PACP from GCP

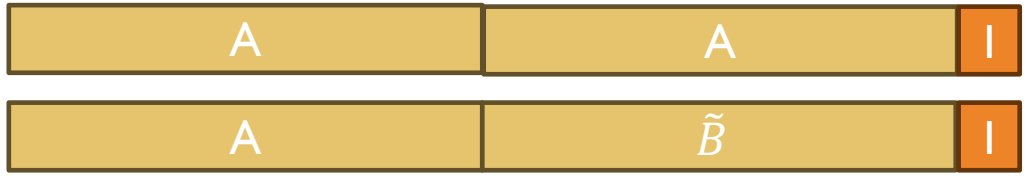
GCP L=26



GCP L=52



PACP L=53



PACP Result (Odd-length)

L = 27
A=101011011110000011110110110
B=111011110011011001100001010

L = 29
A=11111000110101110011010000010
B=11000101011010110010011000000

L = 31
A=1101011101110110110111011100010
B=1110111100101111000101100000000

L = 33
A=000000000000100110101000110100111
B=000011110010001010011100100110101

L =35
A=00000000000011010011001010100011011
B=00000001110001100101011010110010111

L = 27 k=2, C=2=10

A=101011011110000011110110110

PACP Result (Odd-length)

L =37
A=1001101011111100110110010101101110000
B=1110111110101101111000101111000100000

L =39
A= 1, 1, -1, 1, 1, -1, -1, -1, -1, 1, 1, -1, 1, 1, -1, -1, 1, -1, -1, -1, -1, -1, 1, -1, -1, 1, -1, 1, -1, -1, 1, -1, -1, -1, -1, -1, 1, -1, -1
B= 1, 1, -1, 1, 1, -1, -1, -1, -1, 1, 1, -1, 1, 1, -1, -1, 1, -1, -1, -1, -1, -1, 1, -1, -1, 1, -1, 1, -1, -1, 1, -1, -1, -1, -1, -1, 1, -1, -1

L =41
A=10000111000000010111011101100010010100000 B=11100010011101101001101101101010111111000

L =43
A=1110000100010010001000111110001010010010110
B=1010110000100001111001000110010011111011010

L =45
A=101010001100111100000010000001111011100010110
B=100011101001110110100110110100100001010001100

PACP Result (Odd-length)

L =47

A= 1, 1, 1, 1, 1, -1, 1, 1, 1, 1, -1, -1, 1, -1, 1, -1, 1, 1, 1, -1, -1, 1, -1, -1, 1, 1, -1, 1, 1, -1, -1, -1, 1, -1, 1, -1, 1, 1, -1, -1, -1, -1, 1, -1, -1, -1, -1

B = 1, 1, 1, 1, 1, -1, 1, 1, 1, 1, -1, -1, 1, -1, 1, -1, 1, 1, 1, -1, -1, 1, -1, -1, 1, 1, -1, 1, 1, -1, -1, -1, 1, -1, 1, -1, 1, 1, -1, -1, -1, -1, 1, -1, -1, -1, -1

L = 49

A=1011110001111011001010100010001101111101100010110

B=1111011001100111101101001110000000011110010111010

L=51

A=100010100101100011111101101101111100011010110011 10

B=101111110001010111011011110001001000101011110011100

L=53

A= 01001101111010111100111010010110010000111111100111010

B= 01001101111010111100111010101001101111000000011000101

PACP Result (Odd-length)

L=55
A=1100000111001100101011100000000011101010010001110100110
B=1011111011111011000101101000110100101110011000010100110

L=57
A=111011010001101111101100000111000001101111101111011011110
B=101010001110001101011011100110110001001010011111010000010

L=59
A=10000101011101011110110100111111100101101111110110011000110
B=11101100101101111001100000111000011111001100101000010101110

L = 63
A=111101111101010001100011011101101101110110001100010111010010100
B=110010101110000111000000101101110010010111110001111101110101110

PACP Result (Odd-length)

L = 67

A = 1 1 -1 -1 1 -1 1 -1 -1 1 1 -1 -1 -1 1 1 1 1 -1 1 -1 1 1 1 1 1 1 -1 -1 1 -1 -1 -1 1 -1 1 1 1 -1 1 1 -1 -1 -1 -1 -1 -1 1 -1 1 1 -1 -1 -1 -1 1 1 1 -1 -1 1 1 -1 1 -1 1 1 -1

B = 1 -1 -1 1 -1 1 -1 -1 1 1 -1 -1 -1 1 1 1 1 -1 1 -1 1 1 1 1 1 1 -1 -1 1 -1 -1 -1 1 -1 1 1 1 -1 1 1 -1 -1 -1 -1 -1 -1 1 -1 1 -1 -1 -1 -1 1 1 1 -1 -1 1 1 -1 1 -1 1 1 -1

L=107

A = 1 1 -1 1 1 -1 -1 -1 -1 1 1 1 1 1 1 -1 1 -1 -1 1 -1 -1 -1 1 -1 1 -1 1 -1 1 1 -1 -1 1 1 1 1 1 -1 1 1 1 1 -1 1 -1 -1 1 1 1 -1 -1 1 1 -1 -1 1 1 -1 -1 -1 1 1 -1 1 -1 -1 -1 -1 1 -1 -1 -1 -1 -1 1 1 -1 -1 1 -1 1 -1 1 -1 1 1 1 -1 1 1 -1 1 -1 -1 -1 -1 -1 -1 1 1 1 1 -1 -1 1 -1

B = 1 -1 1 1 -1 -1 -1 -1 1 1 1 1 1 1 -1 1 -1 -1 1 -1 -1 -1 1 -1 1 -1 1 -1 1 1 -1 -1 1 1 1 1 1 -1 1 1 1 1 -1 1 -1 -1 1 1 1 -1 -1 1 1 -1 -1 1 1 -1 1 -1 -1 -1 1 -1 -1 -1 -1 -1 1 1 -1 -1 1 -1 1 -1 1 -1 1 1 1 -1 1 1 -1 1 -1 -1 -1 -1 -1 -1 1 1 1 1 -1 -1 1 -1 1

L=907 , 1087 , 10007

PACP Result (Even-length)

L = 28
A=0001001000110001101001010101
B=0000000001100101101100011111

=>非SZCP

L = 36
A=111001100001000010000101000101100101
B=100011111100100010111011010100001101

=>非SZCP

L =38
A=00000000000101110101011010110001100111
B=00001001011100001101000101100010011011

=>非SZCP

L=48
A=111110100110001001000100001000111100000000111010
B=100101011000110010010111000101110100100000110010

=>非SZCP

PQCP Result

L = 44
A=11111101110101100111000011100110110101000011
B=10001001010111110100110011010000001110010010

L = 46
A= 00110110010000101100100001000111010111010010 00
B=00010000100011000001111000010101011011110100 11

THANK YOU

PACP(Odd-length)

27,29,31,33,35,37,39,41,43,45,47,49,51,53,55,57,59,63,
67,71,79,83,103,107,127,151,907,1087,10007,90007

PACP (Even-length)

28,36,38,48

PQCP

44,46